

Quiz1 - Results



Attempt 1 of 1

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Attempt Score 4 / 10 - 40 %

Overall Grade (Highest Attempt) 4 / 10 - 40 %

Question 1

0 / 2 points

Computing $x^y \bmod N$ by repeated squaring, with n the bit length of $\max(x, y, N)$, takes what total time?

- ☒ $O(\log n)$ — the exponent halves at every step
- ☐ $O(2^n)$ — the same as multiplying x by itself $y - 1$ times
- ☐ $O(n^3)$ — about $O(n)$ calls $\times O(n^2)$ per modular multiplication
- ☐ $O(n^2)$ — it is a single modular multiplication

Question 3

0 / 2 points

For “algorithms on numbers” in Week 2, when the input is $N = 1000$, what is the input size n ?

- ☐ $n = 1000$ — the value of the number
- ☒ $n = 4$ — only the decimal digits count
- ☐ $n = 1$ — one number is one input
- ☐ $n \approx \log_2 N \approx 10$ — the number of bits it takes to write N down

Question 4

0 / 2 points

The naïve recursion $\text{fib1}(n) = \text{fib1}(n-1) + \text{fib1}(n-2)$ runs in roughly what time?

- ☐ Logarithmic time, $O(\log n)$
- ✖ ☒ Linear time, $O(n)$
- ➔ ☐ Exponential time — it grows as fast as the Fibonacci numbers themselves (about ϕ^n)
- ☐ Quadratic time, $O(n^2)$

Done